

Preliminary Wrist Angle Estimation Using FBG Sensors for Hand-Exoskeleton Monitoring

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Abstract—Post stroke rehabilitation requires accurate monitoring of joint movements to support recovery and improve functional outcomes. This work presents the design and validation of a soft, wearable sensor based on fiber Bragg grating (FBG) technology for estimating wrist joint angles in the sagittal and frontal planes. For this, two FBGs were inscribed and encapsulated in a viscoelastic material to enhance robustness and sensitivity. Experimental validation was conducted by comparing the sensor measurements with angles derived from a Huskylens camera vision system and a goniometer in controlled trials in both planes. Results after linear regression models implementation showed linearity with R^2 values of 0.96 and 0.94, and root mean square errors of 3.69° and 4.59° for sagittal and frontal movements, respectively. These results demonstrate the feasibility of integrating the proposed FBG sensor into hand exoskeletons for motion tracking in post stroke rehabilitation.

Index Terms—Angle Measurement, Hand Exoskeleton, Optical Sensor, Rehabilitation, Soft Robotics.

I. INTRODUCTION

According to the World Stroke Organization (WSO), stroke has been one of the main causes of death [1], and it is one of the leading causes of disability-adjusted life years worldwide. Consequences of stroke, such as hemiparesis, hemiplegia, sensory deficits, and cognitive impairments, reduce the functional independence of survivors, making rehabilitation an essential component to improve motor functions [2]. These effects also have an impact on hand function.

Loss of hand function limits patients from performing activities of daily living (ADLs) [1]. Good adherence to physical therapies and rehabilitation is essential for a more efficient recovery process. This includes motor skill exercises, mobility training, and active or passive range of motion (RoM) exercises [3]. Rehabilitation is a long-term process that requires a comprehensive approach. Therefore, technologies such as exoskeletons and robotic grasping assistive devices are included [4]. Within these devices has involved soft technologies, which focus on the properties of the materials, including high mechanical compliance, high degrees of freedom, and adaptation to different shapes [5].

In addition, quantitative variables are usually used to monitor the evolution of patients, for which sensing devices

are essential to provide valuable information throughout sessions [6]. These devices allow monitoring of the patient's movements, force exertion, and posture, providing accurate data of performance and therapy effectiveness. Some exoskeletons use encoders, Inertial Measurement Units (IMUs), and piezoelectric sensors [7]. Thus, hand exoskeletons assist in finger movements such as flexion and extension to rebuild hand functions [8]. Moreover, to have an integral insight of the hand function, it is required the monitoring of the wrist movements [9]. Some applications have tracked the wrist flexion, extension, and axial rotation. There are also devices that only support finger rehabilitation based on finger extension training to promote ADLs [10]. However, for soft-based technology, non-rigid sensors are required to measure the system and user's metrics without decreasing its flexibility advantages, and avoiding the adaptation of complex-sensing systems [11].

In this context, optical fiber sensors, especially fiber Bragg grating (FBG) sensors, are considered an alternative solution owing to their small size, high sensitivity, immunity to electromagnetic interference, and multiplexing capability [12]. This type of technology is found in touch applications, gesture recognition, angle measurements, and other sensing fields [13]. In hand exoskeletons, even being applied in some works, exploring finger angles and movements detection [14].

In this way, this work presents the development of an optical fiber sensor based on FBG to measure wrist angles on two axes (sagittal and frontal planes). The sensor validation was performed through a comparison with a vision-based camera. This contribution will allow the integration of the sensor in a soft-exoskeleton for post stroke hand rehabilitation.

II. METHODOLOGY

A. Sensor Working Principle

The FBG working principle is based on a distributed Bragg reflector fabricated in a short segment of the fiber core to reflect a specific wavelength range of the guided modes and transmits all others [15]. The Bragg wavelength λ_B is known as the central wavelength and is calculated as $\lambda_B = 2n_{\text{eff}}\Lambda$, where n_{eff} is the effective refractive index of the fiber core, and Λ is the period length in the periodic variations in the FBG wavelength.

When temperature or strain is applied to the structure, it induces periodic variations and alters the effective refractive index, resulting in a shift of the λ_B . The relationship between the λ_B , the applied temperature (ΔT), and strain ($\Delta\epsilon$) is described in Equation 1.

$$\frac{\Delta\lambda_B}{\lambda_{B0}} = (1 - \rho_\epsilon)\Delta\epsilon + (\alpha + \xi)\Delta T, \quad (1)$$

This work is partially supported by FAPES (EXTRATO ATA DA 6ª REUNIAO CCAF/2022 – “CPID 2.2”, FAPES/ CNPq/Decit – PPSUS 17/2024), CNPq (306323/2024-9, 404111/2023-8, and 402739/2024-8), and IEEE RAS SPARX.

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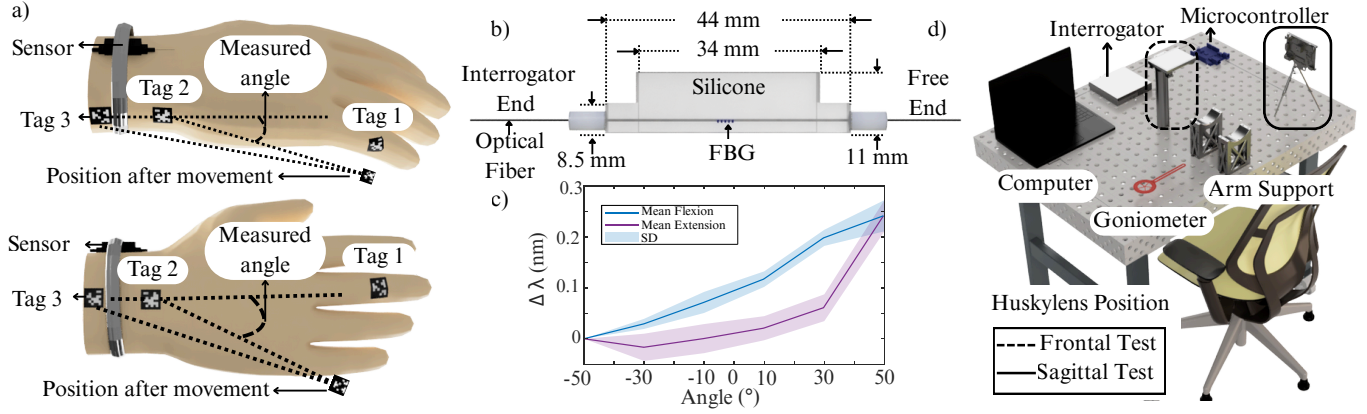


Fig. 1. Proposed sensor and experimental set up. a) Sensor and tag positioning for sagittal test and frontal test, b) Developed FBG-based sensor for angle estimation with its dimensions, c) Hysteresis response and d) Proposed experimental setup with vision-based camera different positions.

where ρ_e , α , and ξ are the photoelastic coefficient, thermal expansion coefficient, and thermos-optic coefficient of the fiber, respectively.

B. Sensor Design

Two FBGs were fabricated using an ultraviolet (UV) laser (CNI DPS-266, China) operating at a wavelength of 266 nm and emitting nanosecond pulses of 8 ns at a repetition rate of 100 Hz. This, combined with a phase mask (Ibsen Photonics, Denmark), inscribes the periodic refractive index modulation along the core of a single-mode optical fiber [16]. For this, a boron-doped photosensitive fiber (FiberCore, United Kingdom) was required, which must have its core exposed to ensure adequate UV exposure. The resulting FBGs were centered at a λ_B of 1582.60 nm.

After obtaining the fiber with the inscription, the mechanical resistance was improved by embedding it in Ecoflex silicone coating (Polisil, Brazil), which not only reinforced the structure but also enhanced the sensor's response to deformation. Additionally, a PTFE tube was attached to each end of the fiber. The design is illustrated in Figure 1b with an hysteresis response of 50.3% in 1c. For testing, the FBG sensor was secured to an user using a polyester and elastodiene support.

C. Experimental setup

The setup was designed to test and compare the right wrist angle measurement in two planes, sagittal (flexion and extension) and frontal (abduction and adduction). It was made to provide a comfortable space for sensor placement and data collection. To avoid fatigue effects due to holding the hand in fixed position, a 3D printed structure was designed to support the forearm and wrist. During the procedure, the user remains seated with the FBG sensor connected to the interrogation system (Hyperion Si255, LUNA, Inc., United States), that uses the ENLIGHT Software (Micron Optics, Inc., United States) on a computer for data record. The researcher controlled the real angle with a goniometer, giving indications for the flow of the session.

For the study, the reference sensor used was the Huskylens Camera (DFRobot, China) with the predetermined function of tag tracking in real time. The Huskylens was connected to an Arduino Uno microcontroller (Arduino, Italy) to collect the

kinematic data and convert it into angle. In addition, for each plane test, the Huskylens was located in a different position, for the frontal plane the tripod was 20 cm aside the arm support, and for the sagittal plane 30 cm above the wrist, as shown in Figure 1d.

D. Procedure

For sensor characterization, two different tests were performed, one for each plane. The test involved performing full-range joint movements: from maximum flexion to maximum extension in the sagittal plane, and from maximum adduction to maximum abduction in the frontal plane. Variations of 10° were made every 10 s while collecting data from both the camera and the sensor. This test was performed by a healthy subject five times for each movement.

For data analysis, the Huskylens camera system delivers the position (x,y) in pixels in its reference system. The position values of the three tags were stored, which changed their position according to the movements of the wrist, as shown in Figure 1a. To obtain the joint angle, the cosine law was applied, and the supplementary angle was calculated, as presented in Equation 2.

$$\theta = 180 - \cos^{-1} \left(\frac{a^2 + c^2 - b^2}{2ac} \right), \quad (2)$$

where a , b , c are the distances between the tags calculated from the coordinates. Due to tag data loss, for the moments when the information of a point was empty, the previously read value was used.

The FBG data were processed using a level-adaptive Discrete Wavelet Transform (DWT) and saved in a time-table format in MATLAB R2024a (MathWorks, USA) for synchronization and segmentation based on the time of each step. The results of each test were labeled and grouped with the angles, where two linear regressions with 5-fold cross-validation were performed, one for each plane's movement.

III. RESULTS

Figure 2a shows the camera values for the five trials for both movements. It can be seen that Tags 2 and 3 tend to maintain a small dispersion, while Tag 1 is responsible for providing the joint angle, as it is in the area of greater displacement. When observing the movement patterns in each

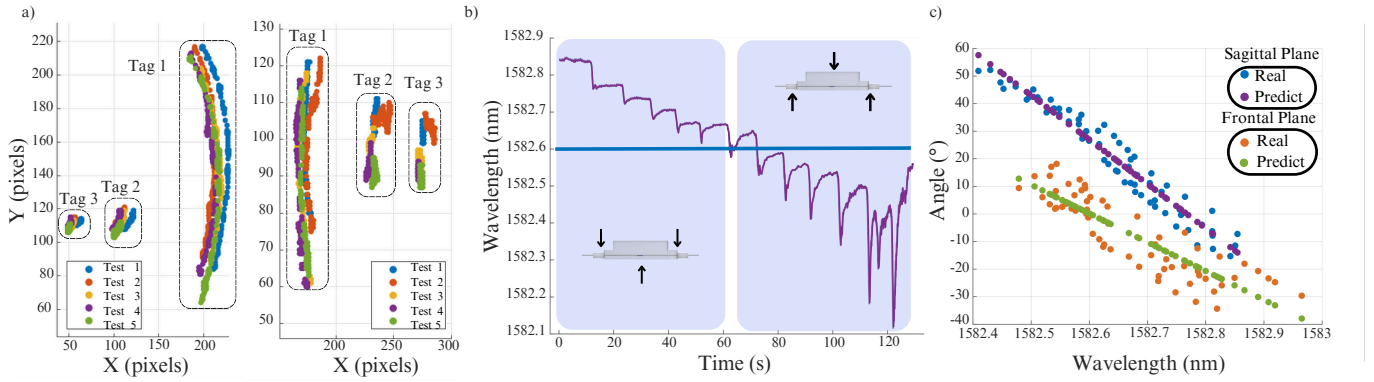


Fig. 2. Camera and sensor response. a) Camera tag tracking system for sagittal and frontal plane, b) FBG sensor response according to the type of movement generated, showing the central value of the Bragg wavelength, c) Linear regression for planes movements showing real data and predicted response.

experiment, the relationship between the maximum angles for flexion ($\sim 60^\circ$) and extension ($\sim 20^\circ$) can be appreciated, with the joint showing greater flexion capability. The same behavior is observed in the frontal plane, with greater adduction ($\sim 40^\circ$) than abduction ($\sim 20^\circ$).

Figure 2b shows a response of the FBG sensor. It can be seen the signal in the form of a step due to changes in the angle of the joint throughout the test. In addition, it can be seen that there are two patterns of sensor response, which are differentiated in the Figure. The behavior is due to the configuration of the silicone cover, where the fiber is not embedded in a symmetrical structure as it is observed in Figure 1b. This type of behavior was observed throughout all tests.

The linear regression obtained to relate the sensor measurements with the joint angles for movements in the sagittal plane is shown in Figure 2c. The tendency of the linear behavior in response to the angle variation can be observed, obtaining an R^2 of 0.96 and an root mean square error (RMSE) of 3.69° . The resulting regression for the sagittal plane is given by $Angle[^\circ] = 254585.625 - 160.848\lambda$. A linear regression was also performed for the data obtained in the frontal plane shown in Figure 2c. In this case, the characteristic equation obtained is presented by $Angle[^\circ] = 165126.066 - 104.338\lambda$, for which an R^2 of 0.94 and an RMSE of 4.59° were obtained.

IV. DISCUSSION

The results obtained from the relationship between the sensor and the measured joint angles demonstrate good performance. For instance, Vogt *et al.* used a soft sensor for the angle measurement and obtained an R^2 of 0.97 for radial/ulnar deviation movements and 0.98 for flexion-extension [17]. In comparison, the results from this study yielded R^2 values of 0.94 and 0.96, respectively. For the study of the wrist, other types of technology can be found such as barometric systems [18], stretch sensors [19], among others. These technologies reported drawbacks for applications where the fingers had a load that alter the measurements of gestures or angles, in the case of the stretch sensors, more advanced processing was required to obtain lower R^2 values than expected.

Moreover, the use of sensors to measure wrist joint angles in post stroke patients offers a practical method for monitoring motor recovery during rehabilitation. Continuous tracking of wrist RoM can provide clinicians with quantitative feedback

to evaluate progress and adapt therapies [20]. Recent evidence indicates that improvements in wrist RoM are associated with enhanced hand function, particularly in tasks involving grasp and object manipulation [21]. In this context, access to real-time angular data supports the assessment of therapeutic effectiveness and contribute to more structured and outcome-oriented rehabilitation strategies. In addition, the proposed sensor could be feasible in post-stroke rehabilitation exercises. Typical wrist exercises are performed at low frequencies (0.25-1 Hz, $<200^\circ/s$), frequencies that are below FBG interrogators (≥ 1 kHz) [22], [23]. On the other hand, the RMS errors obtained (3.69° sagittal, 4.59° frontal) are within the minimum detectable change reported for goniometry (5-10 $^\circ$), representing only 2-3% of the functional arc of flexion-extension [24], [25]. The hysteresis behavior of the sensor is due to the combination of viscoelastic material for protection and elastic for fixation. These values can also be explained by the asymmetrical design of the sensor, which opens up possibilities for future work in which this behavior can be used to determine gestures or classify movements.

Among the limitations of the sensor and its applications, these are mainly related to the nature of the FBGs, which require high-cost interrogation systems and can interfere with implementations where a portable device is required [26].

V. CONCLUSION

It was possible to validate the performance of the FBG-based sensor for angle measurement using a camera system with computer vision. The results obtained from this comparison indicate that the system has a good performance, since were obtained R^2 values for the sagittal and frontal plane of 0.96 and 0.94, respectively. In addition, performing multiple tests represents the repeatability of the system. The size, robustness, and results obtained, make the sensor a candidate for applications in exoskeletons, which enable monitoring users information to generate quantitative feedback in rehabilitation processes. For future work, the use of the sensor for gesture detection during ADLs will be evaluated, with a view toward its application in assistive devices Taking into account anatomical differences among users. New models can also help to extract information about rotational movements when working with simultaneous signals, having a complete mapping of the position and movements of the joint.

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